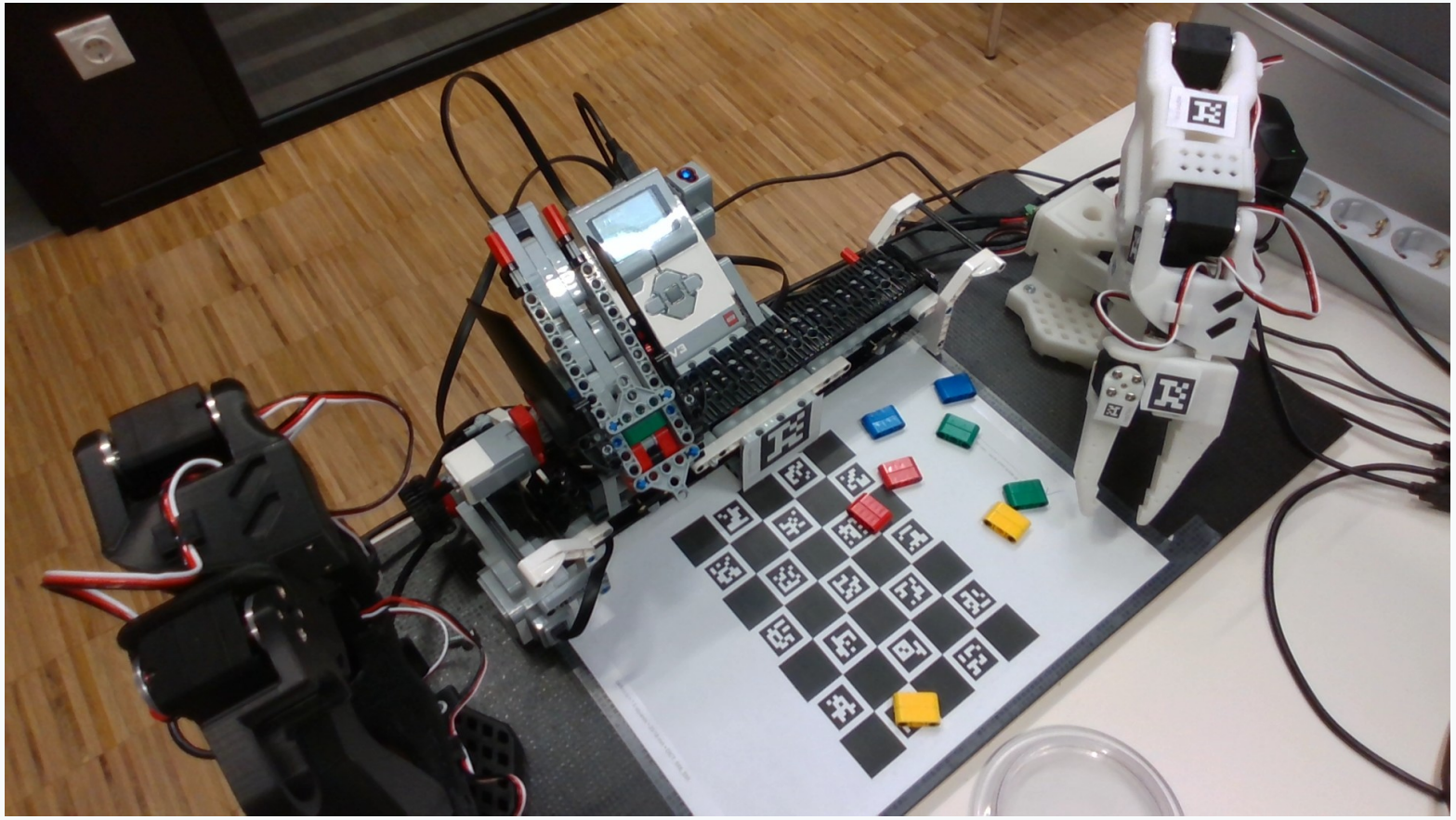
Our fixed-fixture factory



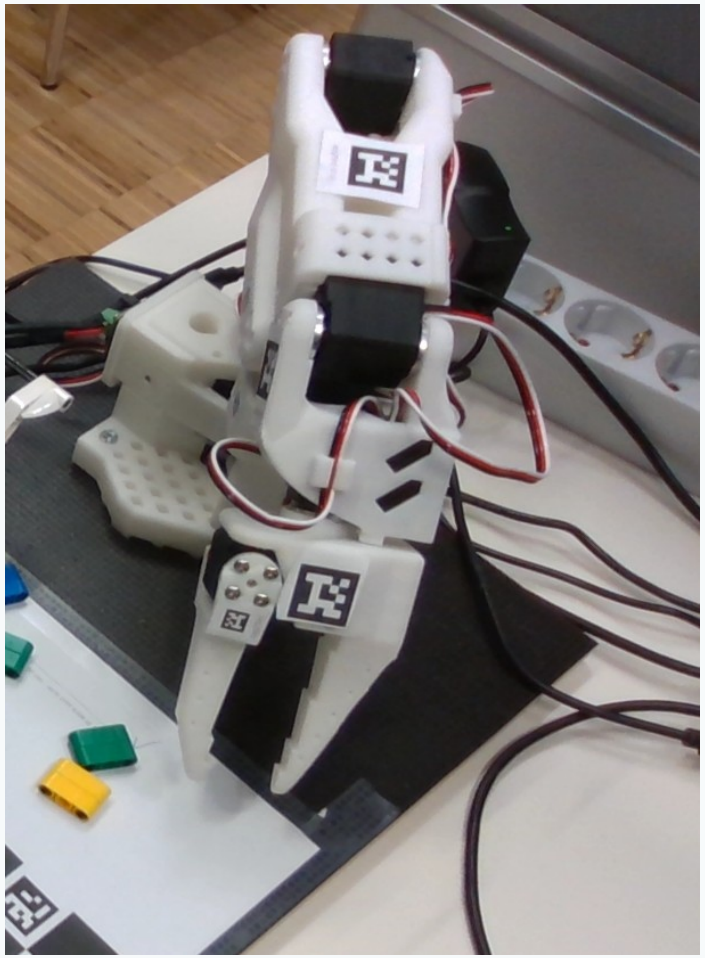
Dispense one object

Carry it to a stopped pickup fixture

Pick, place, release and return

Actual station photograph · one fixed pickup and one fixed destination

Measure decide act check



Sensor measures

↓

Program decides

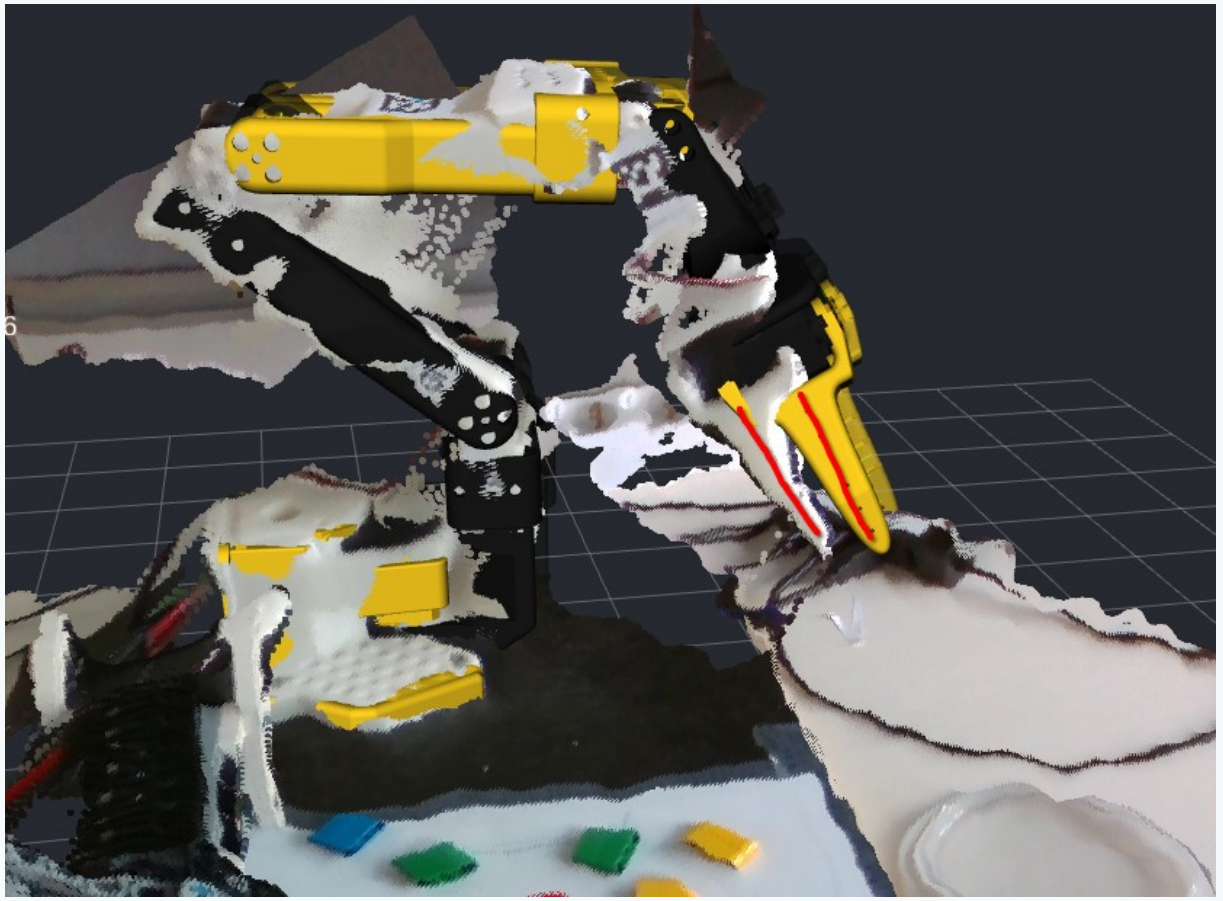
↓

Actuator moves

↓

Inspection checks the result

Robot and screen twin



Leader: handle

Follower: physical arm

Twin: screen model

Put the steps in order

1 Dispense

↓

2 Conveyor to fixture and stop

↓

3 Approach and grasp

↓

4 Lift and place

↓

5 Release and return

An observation can be wrong

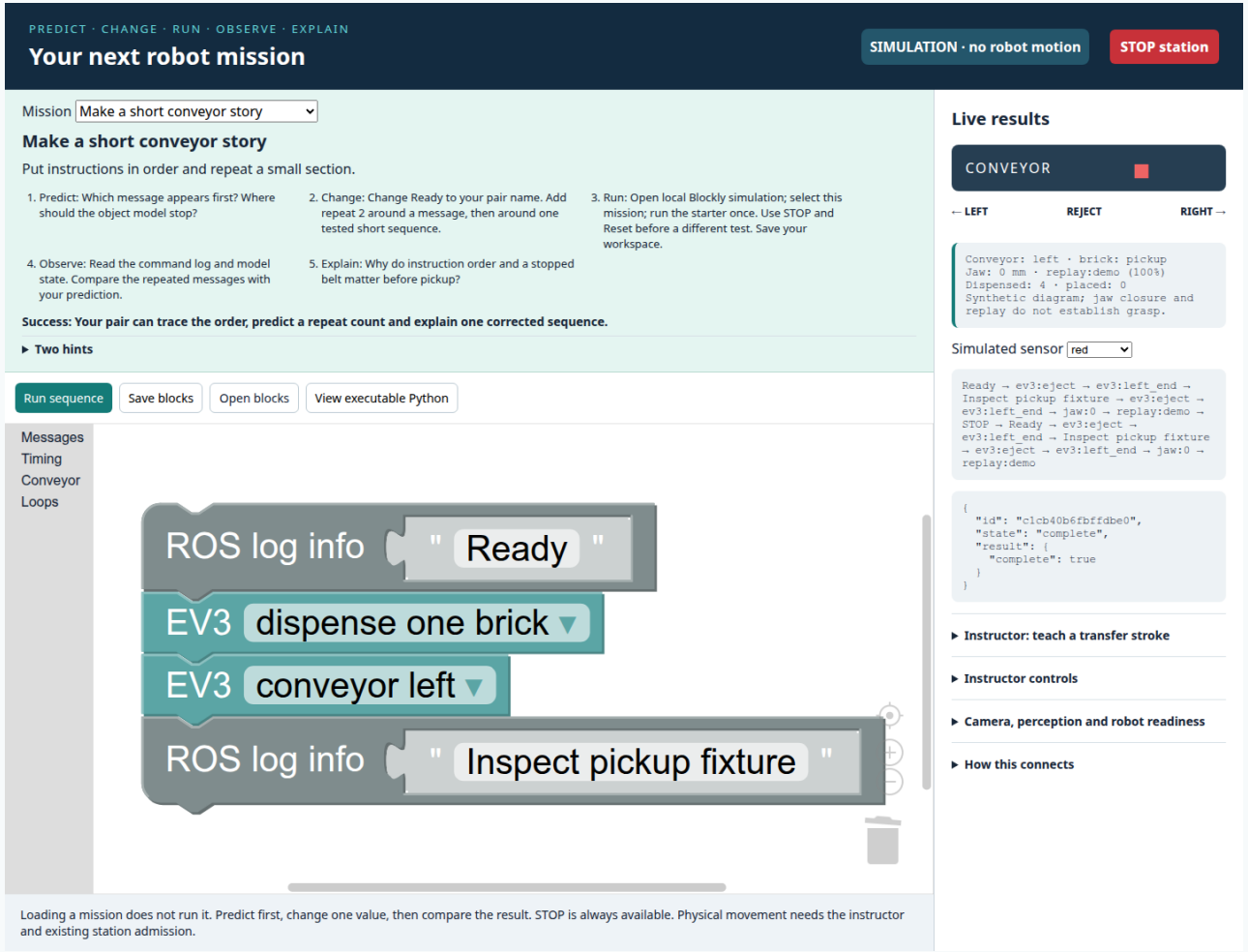


Actual object: red

Reported label: blue (altered example)

Missing report: pause and inspect

Blocks run in order



Read from the top

Repeat the blocks inside

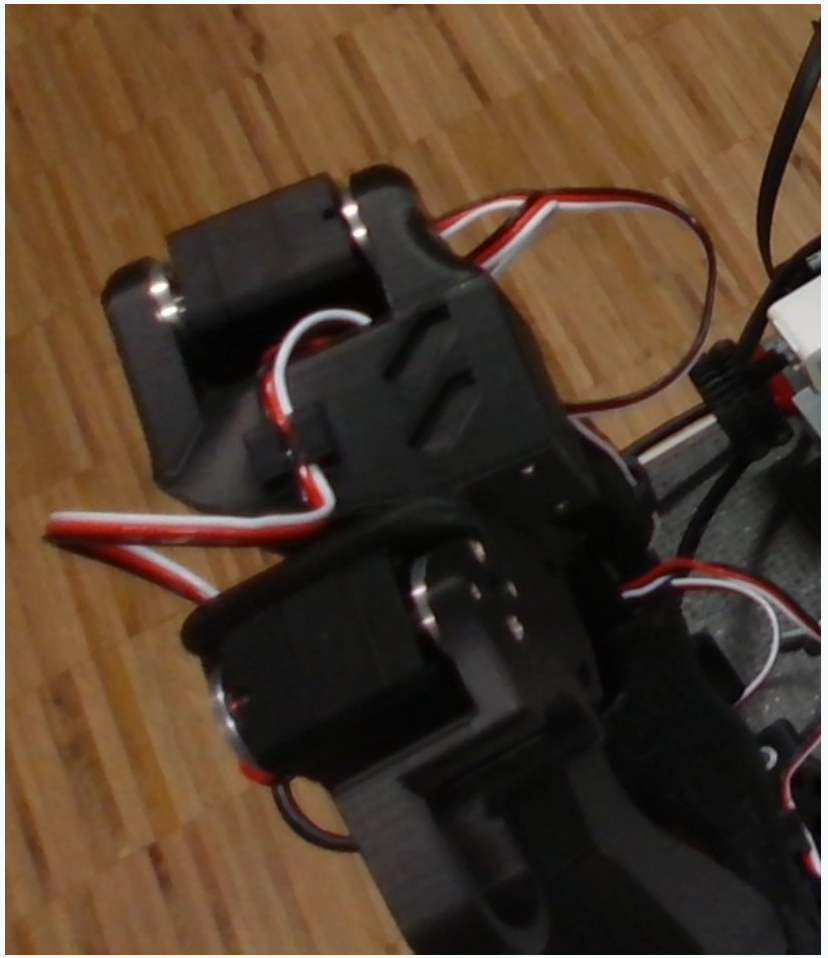
Predict before Run

Your first Python change

```
TEAM = "Pair A"
REPEATS = 3
print("Hello", TEAM)
print("Count", REPEATS)
```

Predict the output, then change one value.

Teach inspect return replay



Teach a short stroke

Inspect and validate

Return to the validated start

Replay separately

Make a short conveyor story

Mission 1 · 25 minutes total · Predict and run

Worked example	Ready → dispense → belt to pickup → inspect. Simulation is a model.
Predict	Which instruction runs first? Where should the model stop?
Change	Change Ready to your pair name. Repeat one tested message twice.
Run	Run the local Blockly starter once. Use Reset simulation between experiments.

Make a short conveyor story

Mission 1 · 25 minutes total · Check and explain

Observe	Compare command order, model position and repeated messages.
Explain	Why must the conveyor stop before the arm approaches?
Success	Trace the order, predict repeat count and explain one correction.
Hints	Read connected blocks from top to bottom. Repeat runs every block inside it.

Optional: Try a wrong order in simulation and explain what changed.

Separate the object from its report

Mission 2 · 20 minutes total · Predict and run

Worked example	Actual blue + missing report: pause. Actual red + blue report: inspect.
Predict	Which report is missing? Which disagrees with the actual object?
Change	Change one report to None; then change a known report to a wrong color.
Run	Run or trace the three rows. Use the supplied labelled photograph.

Separate the object from its report

Mission 2 · 20 minutes total · Check and explain

Observe	Record actual object and reported label in separate columns.
Explain	Why can a fresh, familiar label still be wrong?
Success	Identify one missing and one wrong report; choose pause and inspect.
Hints	Compare actual and reported one row at a time. A known word is still only a report.

Optional: Test green without adding a rule.

Change a name and a number

Mission 3 · 30 minutes total · Predict and run

Worked example	TEAM stores text; REPEATS stores a number. Indentation defines the loop.
Predict	What greeting and how many object lines will appear?
Change	Change TEAM. Run. Then predict and change REPEATS to 2.
Run	Save a pair notebook copy; run from the top with Shift+Enter.

Change a name and a number

Mission 3 · 30 minutes total · Check and explain

Observe	Count lines. Compare program.program command rows with the program list in View executable Python.
Explain	Which edit changed text? Which changed the number of repetitions?
Success	Both partners make and explain one edit; share the same validated commands.
Hints	Quotes surround text, not the repeat number. Keep four spaces before the repeated print.

Optional: Change a message in Python and Blockly; compare both traces.

Watch teach inspect return replay

Mission 4 · 15 minutes total · Predict and run

Worked example	Leader 10°, follower 30°, leader now 15°: relative request 35°.
Predict	Can replay begin from any pose? What if the jaw state changed?
Change	Change one invented angle after predicting. Instructor owns physical modes.
Run	Watch: teach, stop recording, inspect, return to validated start, replay.

Watch teach inspect return replay

Mission 4 · 15 minutes total · Check and explain

Observe	Identify start, endpoint, jaw state and validation evidence.
Explain	Why separate teaching from replay? Why reteach after calibration changes?
Success	Explain teach → inspect → return → replay, and locate STOP.
Hints	A saved stroke does not plan from every start pose. A matching screen does not prove a physical grasp.

Optional: Compare relative 35° with absolute 15°. Invented arithmetic sends no motion.

Explore three stations

Mission 5 · 45 minutes total · Predict and run

Worked example	SIMULATION: changed leader now 10° → 15°; predicted and calculated relative 30 + 15 - 10 = 35°. No physical joint was measured.
Predict	Predict one direction, report or instruction order at this station.
Change	Change one angle, report or message. Swap driver and observer.
Run	15 minutes: 2 predict, 7 change/run, 4 observe/explain, 2 transition.

Explore three stations

Mission 5 · 45 minutes total · Check and explain

Observe	Record one observation and label it PAPER, SIMULATION or PHYSICAL.
Explain	Which part measures, decides or acts? What can your evidence establish?
Success	Every pair visits all three positions and records prediction plus evidence.
Hints	The twin is a model updated from a stream. Use the same short sequence at the offline desk.

Optional: Compare fresh with stale without guessing a missing angle.

Dispense carry pick and place

Mission 6 · 45 minutes total · Predict and run

Worked example	Dispense → conveyor/stop → fixed_pick_place. Missing qualification blocks hardware.
Predict	Where must the object and arm be before each step? What should stop the run?
Change	Reorder paper cards. Use the named skill fixed_pick_place in simulation.
Run	Rotate: supervised integration, offline rehearsal, failure/evidence desk.

Dispense carry pick and place

Mission 6 · 45 minutes total · Check and explain

Observe	Inspect dispense, arrival, grasp, lift, release and return separately.
Explain	Which measurement supports each step? What remains unverified?
Success	Explain the full sequence and label synthetic rehearsal versus observed transfer.
Hints	Stop belt; match the validated start and jaw state. Missing qualification blocks hardware; ask instructor.

Optional: After fixed-fixture success, discuss separately validated color destinations.

Find one failure and test a repair

Mission 7 · 20 minutes total · Predict and run

Worked example	present, missing, present: pause after one inspected simulated object.
Predict	What should happen at the missing middle report?
Change	Choose one failure and one repair. Reset before the repeat check.
Run	Reproduce on paper or in simulation. Repair and rerun the same input.

Find one failure and test a repair

Mission 7 · 20 minutes total · Check and explain

Observe	Record failure message, stopped state and the repaired trace.
Explain	Did the test detect the problem? What could still fail on hardware?
Success	Show a failed prediction, one repair and a repeated check with limits.
Hints	Change one cause so you can explain its effect. One passing run can still hide a wrong known report.

Optional: Try three reds and one blue to expose swapped extension rules.

Show one prediction and result

Mission 8 · 10 minutes total · Predict and run

Worked example	SIMULATION: repeat count 1→2. Predicted Ready twice; log showed Ready twice. This checks messages, not physical arrival or grasp.
Predict	Which saved output best supports your claim?
Change	Choose one change and one observation for the demonstration.
Run	1 minute prepare; 3 × 2-minute pair demos; 3 minutes reflection.

Show one prediction and result

Mission 8 · 10 minutes total · Check and explain

Observe	Show the trace or worksheet; label paper, simulation or physical evidence.
Explain	What next test would increase confidence?
Success	All three pairs demonstrate; both partners speak and name a limitation.
Hints	Read your prediction before showing the result. A trace shows order; arrival needs physical evidence.

Optional: Explain wrong known color versus missing observation.